



A Glossary of Terms in Physical Rehabilitation

accessory joint motions: See *arthrokinematic joint motions*.

active assisted range of motion: Occurs when some degree of muscle activity assists joint range of motion activities.

active range of motion: The motion of a joint that may be achieved by active muscle contraction.

activities of daily living: Activities involved in self-care, communication, and mobility. Examples are climbing stairs, the ability to eliminate independently, and other tasks necessary for normal independent function.

aerobic activity or conditioning: The performance of exercise to increase endurance.

aerobic capacity: A measure of the ability to perform work or participate in activity over time using a body's oxygen uptake, delivery, and energy release mechanisms.

affective: Relating to the expression of emotion; for example, affective disorder.

afferent: Proceeding from the peripheral to the central nervous system.

airway clearance techniques: A broad group of activities used to manage or prevent consequences of acute and chronic lung diseases and impairments, including those associated with surgery.

algometer (pressure): An instrument for measuring the degree of sensitivity to a painful stimulus.

ambulation: Walking, with or without the use of assistive devices.

amplitude: The maximum difference between an alternating current's peak and average values.

anaerobic threshold: The point during exercise at which a person cannot supply enough oxygen to meet the demands of the body.

angiogenesis: Angiogenesis is the physiologic process through which new blood vessels form from preexisting vessels.

anterior: In front; ventral.

approximation: Bringing together two joint surfaces.

arrhythmia: An irregular or abnormal heart rhythm.

arthrokinematic joint motion: Describing the motion of a joint without regard to the forces producing that motion or resulting from it; describing the structure and shape of joint surfaces. Examples of these motions are

glide (slide), roll, spin, distraction or traction, and compression or approximation.

ASIF: Association for the Study of Internal Fixation.

assistive, adaptive, supportive, and protective devices: A variety of implements or equipment used to aid individuals in performing tasks or movements.

atelectasis: Airlessness of the lungs caused by failure of expansion or resorption of air from the alveoli.

athermal: Not using heat, describing, for example, a modality such as pulsed ultrasound.

atlantoaxial luxation: Occurs when instability of the joint allows the axis to luxate dorsally, relative to the atlas, and compress the spinal cord. Congenital in some toy breeds of dogs.

autogenic drainage: Airway clearance through the patient's own efforts (such as coughing).

axonotmesis: The axon integrity is lost, but the endoneurium and Schwann cell sheath it lies within are still intact, providing the opportunity for regeneration back to the correct target. Successful regeneration may occur, particularly if the axon is damaged close to its target.

balance: The ability of an individual to maintain the body in equilibrium with gravity both statically (e.g., while stationary) and dynamically (e.g., while walking).

ballistic stretching: A form of stretching in which a series of quick movements are used to stretch the muscles and connective tissues.

beam nonuniformity ratio (BNR): Regarding therapeutic ultrasound, compares the maximal intensity from the transducer to the average intensity. This ratio should be low (between 2 and 6) to indicate that the energy distribution is relatively uniform.

biofeedback: A training technique that enables an individual to gain some element of voluntary control over muscular or autonomic nervous system functions using a device that produces auditory or visual stimuli.

biomechanical: Describing the action of forces on the body, especially as they affect the musculoskeletal system.

body mechanics: The interrelationships of the muscles and joints as they maintain or adjust posture in response to environmental forces.

carpal hyperextension: A problem of the carpal joints in which the palmar carpal ligaments and palmar carpal

fibrocartilage are damaged and the dog develops a pal-migrade stance.

case management: The coordination of patient care or client activities.

cauda equina syndrome: A progressive neurologic syndrome characterized by lumbar pain, fecal and urinary incontinence, and possible progressive neurologic deficits caused by soft and hard tissue proliferation associated with lumbosacral intervertebral disk disease.

caudal: In animals, a directional term meaning “toward the tail.”

caudal cervical stenotic myelopathy: See *cervical spondylomyelopathy*.

cervical spondylomyelopathy: A disorder caused by abnormal development of the cervical vertebrae resulting in compression of the spinal cord. Relatively common in Great Danes and Doberman pinschers. Sometimes also called caudal cervical stenotic myelopathy or wobbler syndrome.

chondroprotectant: Various compounds that are proposed to have a positive effect on the health and metabolism of chondrocytes and synoviocytes.

cicatrix: Scar; the fibrous tissue replacing the normal tissues destroyed by injury or disease.

circulation: The passage of blood through the heart, blood vessels, organs, and tissues; it also describes the oxygen delivery system.

client: An individual, business, agency, or other organizational entity receiving consultative services.

clinical indications: The patient factors (symptoms, impairments, deficits, and so forth) that suggest that a particular kind of care (examination, intervention) would be appropriate.

closed kinetic chain exercise: Exercise with foot fixed against a resistance.

coaptation device: A bandage, splint, or cast used to immobilize joints and tissues.

coherence: Light waves that are “in phase” with one another.

collagenase: Collagenases are enzymes that break the peptide bonds in collagen.

collimated light: Light with parallel light rays that therefore spreads slowly as it propagates.

compression therapy: Treatment using devices or techniques that decrease the density of a part of the body through the application of pressure.

compressive joint motion: Compressive or approximation accessory movements of joint surfaces.

conduction: Transfer of heat by direct contact between two objects, for example, a hot pack.

conduction velocity: The speed at which electrical energy is transmitted.

consultation: The provision by a physical therapist or veterinarian of a professional, expert opinion or of advice.

continuous passive motion (CPM): The use of a device that allows a joint (such as the knee) to be exercised without the involvement of the patient, often in the early postoperative period.

contracture, joint: Inability to move a joint through its full range of motion.

contrast bath: Alternately immersing the patient or body part in cold and hot water, most appropriate during the early subacute phase of tissue healing.

convection: Transfer of heat using a liquid or gas medium.

cosmesis: A concern in therapeutics, especially in surgical operations, for the appearance of the patient.

cranial: In relation to animals, a directional term indicating “toward the head.”

cranial cruciate ligament (CCL): Analogous to the anterior cruciate ligament in humans.

cranial nerve: One of the 12 paired nerves (such as olfactory and optic) that emerge from or enter the brain.

critical inquiry: The process of applying the principles of scientific methods to read and interpret professional literature; participate in research activities; and analyze patient care outcomes, new concepts, and findings.

cryokinetics: The combination of cryotherapy with motion (passive, active-assisted, active) to facilitate normal, pain-free movement and to promote reduction of edema.

cryotherapy: Therapeutic application of cold (such as ice).

cyanosis: A bluish or purplish discoloration of the skin or mucous membranes caused by a severe oxygen deficiency.

cyclooxygenase (COX): An enzyme necessary for the production of prostaglandins from arachidonic acid. Two forms of the enzyme are recognized, COX-1 and COX-2. COX-2 is produced in response to inflammation.

debridement: Excision of contused and necrotic tissue from the surface of a wound. *Autolytic:* Self-debridement, that is, removal of contused or necrotic tissue through the action of enzymes in the tissue. *Sharp:* Debridement using a sharp instrument.

deficit: A shortfall in amount or quality. *Developmental:* Difference between expected and actual (lower) performance in an aspect of development (e.g., motor). *Receptive:* A shortfall in the skills involving reception (e.g., in vision, in hearing).

degenerative lumbosacral stenosis: A condition in which soft tissue and bony changes, possibly in conjunction with abnormal motion of the lumbosacral joint, impinge on the nerve roots of the cauda equina.

degenerative myelopathy: A poorly understood neurologic condition associated with progressive motor weakness caused by progressive spinal cord degeneration.

developmental delay: The failure to reach expected age-specific performance in one or more areas of development (such as motor or sensory-perceptual).

diagnosis: A label encompassing a cluster of signs and symptoms, syndromes, or categories. It is also the decision reached as a result of the diagnostic process, which is the evaluation of information obtained from the patient examination organized into clusters, syndromes, or categories.

discospondylitis: An infection of the intervertebral disk and adjacent vertebral bodies, resulting in an inflammatory response surrounding the associated spinal cord or nerve roots.

dislocation: A disturbance or disarrangement of the usual relationship of bones as they enter into the formation of a joint.

dispersion: The property by which light is spread out according to its color as it passes through an object.

distraction joint motion: Distraction or traction motions are tensile (pulling apart) movements between bones.

divergence: The beam divergence of a laser beam is a measure for how fast the beam expands. Beams with very small divergence (i.e., with approximately constant beam radius over significant propagation distances) are called *collimated beams*.

dolorimeter: A device to measure pain.

dorsal: In animals, a directional term meaning “toward the spine.”

dressings: A material (such as a topical agent or gauze) applied to a lesion.

duty cycle: Refers to the percentage of time that ultrasound is emitted during one pulse period. Duty cycle = pulse duration (time on)/the pulse period (time on + time off).

dynamometry: Measuring the degree of muscular power.

dyspnea: Shortness of breath; subjective difficulty or distress in breathing frequently manifested by rapid, shallow breaths; usually associated with serious disease of the heart or lungs.

edema: An accumulation of fluid, often occurring as part of the inflammatory process after trauma.

efferent: Sending information away from the central nervous system.

effleurage: A French word meaning “to skim” or “to touch lightly on,” referring to a series of massage strokes used in Swedish massage to warm up the muscle before deep-tissue work using petrissage.

effusion: The escape of fluid into a body part or tissue.

electrical device: An instrument or modality that applies electrical current to biologic tissue for pain control, tissue healing, or muscle dysfunction; an instrument that records electrical activity from excitable tissues of

the body for purposes of neuromuscular diagnosis, education, or relaxation.

electrical stimulation: Treatment through the application of electricity. *Functional:* The application of electrical stimulation to particular peripheral nerves to allow paretic and paralyzed muscles to make functional and purposeful movements.

electrogoniometry: The measurement of the movement of a joint using an electrical potentiometer.

electromyography (EMG): The recording of the electrical activity of a muscle.

electrophysiologic: Concerned with the electrical activity of various body tissues or systems.

electrophysiologic testing: The process of examining the relationships of body functions to electrical phenomena, such as the effects of electrical stimulation on the tissues, the production of electrical currents by organs and tissues, and the therapeutic use of electrical current.

electrotherapeutic modalities: A broad group of therapeutic physical agents (e.g., neuromuscular electrical stimulation, iontophoresis).

endurance: The ability to perform work over time.

environmental, home, and work barriers: The physical impediments that keep individuals from functioning optimally in their surroundings, including safety hazards (such as throw rugs or slippery surfaces), access problems (such as narrow doors or high steps), and home design (such as a multiple-story environment).

evaluation: A dynamic process in which the physical therapist makes clinical judgments based on data gathered during the examination.

evaporation: The change of a liquid into a vapor at a temperature below the boiling point. Evaporation takes place at the surface of a liquid, where molecules with the highest kinetic energy are able to escape. When this happens, the average kinetic energy of the liquid is lowered, and its temperature decreases.

evoked potentials: The electrical signals recorded from a sensory receptor, nerve, muscle, and so forth, of the central nervous system that has been stimulated, most often by electricity (e.g., auditory evoked potentials).

examination: The process of obtaining a patient history, performing relevant systems reviews, and selecting and administering specific tests and measures.

excursion: Movement within the body, with return to the original state implied (e.g., excursion of the diaphragm).

extracapsular suture stabilization: A procedure used to stabilize cranial cruciate deficient stifles using sutures of various materials to span from the caudal aspect of the stifle to the tibial crest.

exudation: The process of expressing material through a wound, usually characterized as oozing.

fibrocartilaginous embolic myelopathy (FCE): A poorly understood phenomenon in which fragments of intervertebral disk material embolize and create regional vascular compromise to the spinal cord.

fibrosis: Fibrosis is the formation of excess fibrous connective tissue in an organ or tissue in a reparative or reactive process.

fibrotic myopathy: Fibrosis and contracture of skeletal muscle secondary to trauma.

field trails: A sporting event in which dogs compete in hunting, retrieving, or pointing competition.

fluidotherapy: “Dry whirlpool”; the application of dry heat through a fluidotherapy machine.

force plate: A plate embedded in the floor used to measure the force that an individual exerts when walking.

fragmented medial coronoid process (FCP): A condition in which the medial coronoid process of the proximal ulna becomes fragmented, possibly because of elbow incongruity or osteochondrosis.

frequency: The number of cycles per second, expressed as hertz (Hz). Ultrasound and neuromuscular electrical stimulation units generally have various frequency settings.

function: The special, normal, or proper action of any part or organ; the action specifically for which an individual or thing is fitted or employed; an act, process, or series of processes that serve a purpose; to perform an activity or to work properly or normally.

functional limitation: A restriction of the ability to perform a physical action, activity, or task in a typically expected, efficient, or competent manner.

gait: The manner in which a dog walks, characterized by rhythm, cadence, step, stride, and speed.

glide joint motion: Shear or sliding motions of opposing articular surfaces.

goal: The long-term statement or statements that define the patient’s expected level of performance at the end of the rehabilitation process; the functional outcomes of therapy, indicating the amount of independence, supervision, or assistance required and the equipment or environmental adaptation necessary to ensure adequate performance. Desired outcomes may be stated as long-term or short-term as determined by the needs of the patient and the setting.

goniometry manual: The measurement of the movement of a joint by manual methods. *Electrical:* See *electrogoniometry*.

graded forces: A term used in manual therapy to denote the application by the physical therapist of varying amounts of pressure on the patient’s body.

Hansen’s type I disk: An acute explosive intervertebral disk herniation.

Hansen’s type II disk: A chronic progressive intervertebral disk herniation.

herniated disk: The protrusion of one of the spinal disks into an opening in the spinal cord, thereby compressing the nerve root.

Hertz (Hz): Cycles per second.

history: An account of past and present health status that includes the identification of complaints and provides the initial source of information about the patient. The history also suggests the patient’s ability to benefit from physical therapy services.

home, environmental, or architectural barriers: The physical impediments (such as stairs or slippery surfaces) that restrain or obstruct an individual’s ability to function in the usual environment.

Hubbard tank: A shallow tank made of stainless steel, Plexiglas, or tile for administering hydrotherapy.

hydrodynamic: Concerned with the flow of liquids.

hydrotherapy: Treatment with external water.

hygroma: Seroma, hematoma associated with the olecranon.

hypermobility: Abnormally high movement or ability to move (as of a joint).

hypertrophic osteodystrophy: A developmental disease of rapidly growing large- and giant-breed dogs characterized by metaphyseal new bone formation.

hypomobility: Abnormally low movement or ability to move (as of a joint).

impairment: A loss or abnormality of physiologic, psychological, or anatomic structure or function.

infrared heat: A therapy using thermal radiation with a wavelength greater than that of the red end of the visible spectrum.

infraspinatus contracture: Fibrotic contracture of the infraspinatus muscle causing a gait abnormality. Contracture of the infraspinatus muscle limits full extension of the shoulder and causes the affected forelimb to be abducted.

innervation: The supply of nerve fibers to a part of the body, such as an organ.

integrity: The characteristic of being whole or fully functional.

intensity: Refers to the rate of energy delivery per unit area.

intervention: The purposeful and skilled interaction of the physical therapist with the patient, using various methods and techniques to produce changes in the patient’s condition.

intervertebral disk disease: A common condition, especially in chondrodystrophic breeds of dogs, in which protrusion or extrusion of an intervertebral disk causes pain, weakness, and varying degrees of neurologic dysfunction.

iontophoresis: Use of continuous direct electrical current to enhance the transdermal administration of medication.

ischemia: Local anemia caused by mechanical obstruction (mainly arterial narrowing) of the blood supply.

jack: Local irritation, seroma, or chronic periostitis associated with mechanical interference between the forelimbs and hindlimbs.

joint integrity: The conformance of the joints to expected anatomic, biomechanic, and kinematic norms.

joint mobility: The ability to move a joint; takes into account the structure and shape of the joint surface as well as characteristics of tissue surrounding the joint.

kinematic: Having to do with the possible motions of a part or all of the human body.

kinesthesia: The awareness of the body's or a body part's movement.

knocked-up toe: Abnormal carriage of the toes caused by avulsion or rupture of the deep digital flexor tendon.

laser: Short for light amplification by stimulated emission of radiation. A device that creates and amplifies electromagnetic radiation of a specific frequency through the process of stimulated emission. The radiation emitted by a laser consists of a coherent beam of photons, all in phase and having the same polarization.

lateral splint: Splint fashioned from casting material applied to the lateral aspect of the leg.

laxity: Looseness, as in laxity of joint.

lipoxxygenase (LOX): An enzyme that converts arachidonic acid to leukotrienes.

loading: The force placed on a body part (such as a foot or the feet); used often in describing the employment of an assistive device.

lure coursing: A sporting event in which sight hounds chase lures pulled by a rope or cord.

lymphatic: Concerned with the lymph nodes and vessels, which form a system for collecting fluid from the tissues and adding it to the venous blood system.

maceration: Softening by the action of a liquid.

magnetic fields energy, pulsed: A therapy using the intermittent application of energy produced by magnetic fields.

manipulation: A therapeutic movement, usually of small amplitude, accomplished at the end of the available range of motion but with the anatomic range at a speed over which the patient has no control.

manual therapy: A broad group of skilled hand movements used by the physical therapist to mobilize soft tissues and joints for the purpose of modulating pain, increasing range of motion, and so forth.

mechanical: Caused by or derived from machinery; habitual, routine, automatic; related to, controlled, or affected by physical forces (such as a traction device). *Biomechanical:* The physical structure, forces, and movements in the human body.

medial buttress: Periarticular fibrosis and swelling of the medial aspect of the distal femur associated with chronic stifle instability and cranial cruciate deficiency.

meniscal release: Involves cutting the medial meniscus just caudal to the medial collateral ligament. Used in conjunction with the tibial plateau leveling osteotomy procedure.

mentation: A mechanism of thought or mental activity.

metabolic: Concerned with metabolism, the sum of all physical and chemical changes that take place within an organism; all energy and material transformations that take place within living cells.

microvolt: One millionth of a volt.

mobilization: A therapeutic movement accomplished within the available range of motion at a speed that the patient cannot control.

modality(ies): Physical agent(s), including, but not limited to, thermal, acoustic, light, mechanical, or electrical energy, applied to produce therapeutic changes in biologic tissue.

monkey muscle: Long head of the triceps.

motor function: The ability to learn or demonstrate the skillful and efficient assumption, maintenance, modification, and control of voluntary postures and movement patterns. *Fine:* Refers to relatively delicate movements. *Gross:* Refers to larger-scale movements.

multisegment motion: Simultaneous movement of several parts of the body.

muscle length: The length of the muscle during various stages of tension (from resting at full extension through the contractile range); in conjunction with joint integrity and connective tissue extensibility, muscle length determines flexibility.

muscle performance: The capacity of a muscle to do work (force $\times$ distance).

nerve root compression: A squeezing of one of two bundles of nerve fiber emerging from the spine; frequently caused by a herniated disk.

nervous system: The brain, spinal cord, nerves, and ganglia. *Central nervous system:* The brain and spinal cord. *Peripheral nervous system:* The system of nerves in the extremities.

neural: Having to do with a nerve or nerves.

neurapraxia: Axonal conduction is lost without disruption of the axon. This injury usually results from compression, transient ischemia, or blunt trauma. Loss of conduction may be a result of myelin damage or insufficient energy to maintain axonal resting potential.

neuromotor development: The acquisition and evolution of movement skills throughout the life span.

neurotmesis: The entire structure of the nerve is disrupted. The axon has the ability to regenerate but needs to find a Schwann cell sheath to do so, making it much more difficult.

nonvolitional: Involuntary, not controllable.

nutraceutical: Nondrug substances that are produced in a purified or extracted form and are administered

orally to provide compounds required for normal body structure and function with the intent of improving health and well being.

objective: A measurable behavioral statement of an expected response or outcome; something worked toward or striven for; a statement of direction or desired achievement that guides actions and activities.

open kinetic chain exercises: Exercises with the foot not fixed against a resistance.

orthosis: A device (such as a splint or brace) that supports weak or ineffective joints or muscles.

osteocondrosis: A growth-related disease of the cartilage. The primary problem is failure of endochondral bone formation that allows the articular cartilage to irregularly develop and often tear. A cartilage flap is referred to as *osteocondrosis dissecans*.

osteokinematic joint motion: Primary motion of a joint and the movement of bones as a whole, such as occurs with stifle flexion and extension.

osteoporotic: Pertaining to or characterized by a porous condition of the bones; refers to a reduction in the quantity of bone or atrophying skeletal tissue.

oxygen consumption: The amount of oxygen inspired minus the amount of oxygen exhaled.

oxygen saturation: The degree to which oxygen is present in a particular substance.

outcome: The result of physical therapy management expressed in five areas: prevention and management of symptom manifestation, consequences of disease (impairment, disability, or role limitation), cost-benefit analysis, health-related quality of life, and client satisfaction. A successful outcome includes improved or maintained physical function when possible, slows functional decline when the status quo cannot be maintained, or is considered meaningful by the client.

outcomes analysis: A systematic examination of patient outcomes in relation to selected patient variables (e.g., age, sex, diagnosis, interventions performed); outcomes analysis may be used in quality assessment, economic analysis of practice, and so forth.

pain: A disturbed sensation causing suffering or distress.

palpation: Examination using the hands (such as palpation of the spleen).

panosteitis: A growth-related problem of immature rapidly growing dogs. It is characterized by intramedullary deposition of bone and a shifting leg lameness.

paraffin bath: A superficial heat treatment using paraffin wax and mineral oil.

passive range of motion: Motion across a joint with no muscle contraction.

pathomechanical: Describing a disturbance in function not resulting from disease.

pathophysiologic: Describing the functional changes that accompany a particular disease or syndrome.

pathway: A conduction route for nerve impulses.
Sensory pathway: A conduction route for nerve impulses from the sense organs.

patient: One who is being treated for an illness or injury; an individual receiving health care.

percussion (mechanical): A diagnostic procedure in which the clinician taps a body part with a finger or rubber-headed hammer to estimate its density.

performance battery: A set of tests designed to measure a patient's ability to function in a particular area(s).

peripheral circulation: The movement of blood through the extremities.

peripheral vascular: Concerned with the blood vessels of the extremities.

petrissage: Massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling, and pick-up-and-squeeze are the petrissage movements.

phonophoresis: The use of ultrasound to enhance the delivery of topically applied medications through the skin.

photosensitivity: Sensitivity of the skin to light, usually due to the action of certain drugs (or plants, or other substances).

phototherapy: Treatment using the application of light.
Ultraviolet: Light therapy using rays with wavelengths beyond the violet end of the visible spectrum.

physical agent: A form of mechanical, radiant, thermal, acoustic, or electrical energy that is applied to biologic tissues in a systematic manner to achieve a therapeutic effect; a therapeutic modality used to treat physical problems.

physical function: The measurement of physiologic, biomechanical, social, and psychological performance in practical or goal-oriented terms.

physical therapist: A licensed health professional who offers services designed to preserve, develop, and restore maximum physical function.

physical therapist assistant: An educated health care provider who performs physical therapy procedures and related tasks that have been selected and delegated by the supervising physical therapist.

physical therapy aide: A nonlicensed worker, trained under the direction of a physical therapist, who performs designated routine physical therapy tasks.

physiologic joint motion: See *osteokinematic motion*.

planes (midline and segmental): Imaginary flat surfaces drawn through the body; the midline plane bisects the body vertically, whereas segmental planes are drawn at various angles.

posterior: Relating to the back or dorsal side.

postural drainage: Placing the body in a position that causes fluid to drain from the lungs.

postural reactions: The adjustments of the body to gravity required for normal performance; the ability to alter the position of the head, trunk, and extremities to balance one's body with gravity.

posture: The alignment and positioning of the body in relation to gravity, center of mass, and basis of support.

posturography: Procedures to test standing posture, balance, and equilibrium sense.

power: Work produced per unit of time.

presenting problem: The specific dysfunction that causes a client's owner to seek attention or intervention (that is, the chief complaint).

proactive: Seizing the initiative; responding actively rather than passively; performing an action with the idea of influencing events.

prognosis: The determination of the level of maximal improvement that might be attained by the patient and the amount of time needed to reach that level.

proprioception: The reception of stimuli from within the body (such as from muscles or tendons); includes position sense (the awareness of the joints at rest) and kinesthesia (the awareness of movement).

prosthesis: An artificial device, often mechanical or electrical, used to replace a missing part of the body.

pulses: The dilations of arteries (occasionally veins or vascular organs) that correspond to the beating of the heart.

quadriceps tiedown: A disability characterized by an inability to fully flex or extend the stifle following a fracture of the femur.

radiation: Streams of photons, electrons, small nuclei, or other particles. Radiation is given off by a wide variety of processes, such as thermal activity, nuclear reactions (as in fission), and by radioactive decay.

range of motion: Describes the space, distance, or angle through which a patient can move a joint or series of joints.

recurvatum: An angular limb deformity in which the affected joint is excessively hyperextended. The most common form is genu recurvatum, in which the stifle joint is abnormally hyperextended.

reepithelialization: Skin growth to replace skin loss caused by a wound or other injury.

referral: A recommendation that a patient seek service from another health care provider or resource.

reflex: A stereotyped reaction to a variety of sensory stimuli.

reflex inhibition: A situation in which sensory stimuli decrease reflex activity or the voluntary activation of muscle.

rehabilitative: Concerned with restoration of a patient to full or at least improved function.

remediation: The act or process of providing some degree of relief for a patient's clinical problems.

respiration: A term that refers primarily to the exchange of oxygen and carbon dioxide across a membrane into and out of both the lungs and cells.

respiratory quotient: The ratio of the carbon dioxide that the body tissues give off to the amount of oxygen that they absorb.

righting: Adjusting or restoring the body to a desired position.

Robert Jones bandage: A soft, padded bandage to limit tissue swelling and joint motion.

Robinson sling: A bandage used to allow only toe touching of the rear leg. Sometimes called a *Humane Society bandage*.

roll joint motion: Involves one bone rolling on another, such as the femoral condyles rolling on the tibial plateau.

Schroeder-Thomas splint: A splint consisting of a metal hoop and stirrups used to immobilize fractures.

screening: Determining the need for further examination or consultation by a physical therapist or for referral to another health professional.

sensory: Having to do with sensations or the senses; includes peripheral sensory processing (such as sensitivity to touch) and cortical sensory processing (such as two-point and sharp/dull discrimination).

sensory integration: The ability to integrate information from the environment to produce normal movement outputs.

sequential casting: A process in which the patient is casted several times, with each cast less restrictive than the previous one.

sequelae: Aftereffects of a disease or injury.

serosanguineous: Containing both serum and blood.

serous: Like serum, watery.

somatosensory: Having to do with the sensations received in the skin and deep tissues.

somatosensory deficit: A shortfall in the reception of sensations in the skin and deep tissues.

spatial average intensity: In reference to therapeutic ultrasound, the total output (in watts) of the transducer divided by the transducer surface area (in cm²).

spatial peak intensity: The greatest intensity anywhere within the therapeutic ultrasound beam.

spin joint motion: Joint surface motions that result in continual contact of a single area of articular cartilage on adjacent articular cartilage within a joint.

spinal curve: An abnormal curvature of the spinal column (e.g., S curve, kyphosis).

splinting, dynamic: Functional splinting that aids in the movements initiated by the patient and/or controls the plane and range of motion.

spoon splint: A plastic or metal splint shaped like a spoon used to immobilize or support the limb.

sprain: A joint injury without dislocation or fracture involving possible ligament or tendon rupture.

stopper bone: Accessory carpal bone.

standing table: See *tilt table*.

strain: Injury from overuse or improper use.

strengthening, active assistive: A form of strength-building exercise in which the physical therapist applies resistance through the range of motion of the patient's active movement.

strengthening, resistive: Any form of active exercise in which a dynamic or static muscular contraction is resisted by an outside force. The external force may be applied manually or mechanically.

sympathetic disturbance: A malfunction in the sympathetic part of the autonomic nervous system, which governs smooth muscle and the contraction of blood vessels.

symptom magnification scale: An examination tool used to elicit descriptions of levels of pain.

syndrome: The aggregate of signs and symptoms associated with any morbid process that together constitute the picture of a known disease.

synergy: The capability of properly grouping movements in order to perform acts that require special adjustments.

systems review: A brief or limited examination that provides additional information about the patient's general health to help the physical therapist formulate a diagnosis and select an intervention program.

telemetry: The science of measuring a quantity, transmitting the results to a distant station, and then interpreting, indicating, and recording the results.

TENS: See transcutaneous electrical nerve stimulation.

tests and measures: General methods and techniques used to conduct an examination.

therapeutic exercise: A wide range of activities designed to increase strength, improve cardiovascular fitness, increase flexibility, enlarge range of motion, or otherwise increase the body's functional capacity.

thermal: Using heat, as in a thermal agent, for its therapeutic effects.

thermistor: A device for determining temperature; may be extremely small and may also be used to establish and maintain temperature.

thermography: A process of measuring temperature by means of a registering thermometer, one form of which records every temperature variation and registers its rise and fall on a circular temperature chart turned by clockwork.

thermotherapy: The therapeutic application of heat.

tibial plateau leveling osteotomy (TPLO): A corrective osteotomy technique used in the treatment of cranial cruciate ligament disease.

tilt table or standing table: Two kinds of tables used to bring patients from a supine to a vertical position in a deliberate manner.

tissue: Collection of similar cells and the intercellular substances that surround them.

topical agent: An ointment, medication, or similar material applied to the skin for its therapeutic effect.

torque: A force that produces rotation or twisting of a part on its axis.

traction: The therapeutic use of tension created by a pulling force. *Mechanical:* The use of tractive forces to produce a combination of distraction and gliding to relieve discomfort and increase tissue flexibility; also called *passive mobilization*.

transcutaneous electrical nerve stimulation (TENS): The use of electrical current to stimulate cutaneous and peripheral nerves via electrodes on the skin's surface.

treatment: One or more interventions used to cure or ameliorate a disease or pathologic condition or otherwise produce changes in the patient's health status; the sum of the therapies offered to a patient during a complete episode of care.

triage: An initial review of a patient or prospective patient to determine the need for further treatment.

triple pelvic osteotomy (TPO): A surgical procedure involving osteotomies of the ilium, ischium, and pubis and rotation of the acetabulum to increase dorsal coverage of the head of the femur for the treatment of hip dysplasia.

turgor: Fullness, swelling.

ulcer: A break in the skin surface or in a mucous membrane with loss of tissue, usually accompanied by inflammation. *Decubitus ulcer:* Bedsore. *Vascular insufficiency ulcers:* Lesions caused by occlusion of a blood vessel or other vascular disorder.

ultrasound: A diagnostic or therapeutic technique using high-frequency sound waves. Used therapeutically, ultrasound produces heat. *Pulsed ultrasound:* The application of therapeutic ultrasound at frequent predetermined levels.

vasopneumatic compression device: A device intended to decrease swelling by "milking" fluid away from an area, such as an inflatable sleeve strapped around a patient's swollen extremity.

Velpeau sling: A bandage used to secure the forelimb to the thorax, used for immobilization of the shoulder.

ventilation: The movement of a volume of gas into and out of the lungs.

ventral: In animals, a directional term meaning "toward the belly."

vestibular: Describing the sense of balance located in the inner ear.

vital signs: Heart rate, blood pressure, temperature, and respiration rate.

volitional: Intentional, as in controlled movement.

volumetric displacement: The amount of a fluid that leaves a container (of any size) following the introduction of a part or all of the body.

wellness: A concept that embraces a proactive, positive approach to good health. Wellness advocates seek to increase a person's level of health as a preventative measure to guard against future disease.

Wobbler syndrome: See *cervical spondylomyelopathy*.

work conditioning: An intensive, goal-oriented treatment program designed specifically to restore an individual's systemic neuromusculoskeletal functions (strength, endurance, movement, flexibility, and motor control) and cardiopulmonary functions.

zoonosis: The transmission of a disease from an animal or nonhuman species to humans.